Motivation — HackerFab IITB

Hello, I am Kusumuru Suresh Kumar. I am pursuing an M.Tech in Systems and Control Engineering at IIT Bombay and have been building my skills in digital design, verification, and electronic design automation. My VLSI work has taught me to turn a specification into a design, test assumptions, and debug a system using measurable evidence. I now want to understand the physical side of semiconductor technology: how materials, fabrication steps, thermal history, and interfaces determine the behaviour of real devices.

In the short term, I want to contribute to a practical, well-documented hardware project that solves a real process need for a small semiconductor-fabrication team. For this submission, I explored a temperature-controlled hot plate for 4-inch wafer processing. The design connects thermal modelling, mechanical design, sensor calibration, feedback control, safety engineering, and experimental validation. I would begin with a safe prototype and a transparent test plan rather than claim that tight uniformity or repeatability can be achieved from calculations alone.

My proposed approach uses a thick heat-spreading plate, independently controlled centre and edge heater zones, calibrated four-wire temperature sensors, and an independent hardware over-temperature cut-out. The important engineering question is not simply whether the plate can reach 400 °C; it is whether the temperature is accurate, spatially uniform, repeatable from run to run, and mechanically stable after thermal cycling. I would document the assumptions, build a measurement setup, publish raw data, and use the results to decide which parts of the design need to change.

Over the first several months, I would like to learn the team's actual process constraints and safety practices, review the design with people who have hands-on fabrication experience, and help validate one well-defined tool or process. I would prioritise reproducibility: clear drawings, a parts list, assembly notes, calibration procedures, known limitations, and results that another team member can reproduce.

In the longer term, I want to help make semiconductor prototyping more accessible by contributing to low-cost tools and open, carefully documented process knowledge. My experience with Python, digital verification, and structured testing can support data logging, analysis, control experiments, and repeatable validation. I also want to learn directly from fabrication work, because simulation and RTL design do not replace the judgement required to handle real materials, equipment, contamination, and process variation.

I would value the opportunity to learn from the team, take responsibility for a concrete engineering task, and improve it through measured iteration. I see HackerFab as a place to connect design and theory with physical implementation, while contributing documentation and tools that are useful to the wider team